

AS[®]Tar patient case

A clinical impact comparison to standard of care.

A patient case of AS[®]Tar-driven therapy adjustment >18 hours earlier than standard of care.



Patient characteristics

Age: 60

Sex: Male

Clinical history: Dementia or other neurological conditions, arterial hypertension

Overview

Admitted to the ICU due to intracerebral haemorrhage. He developed a BSI of unknown origin.

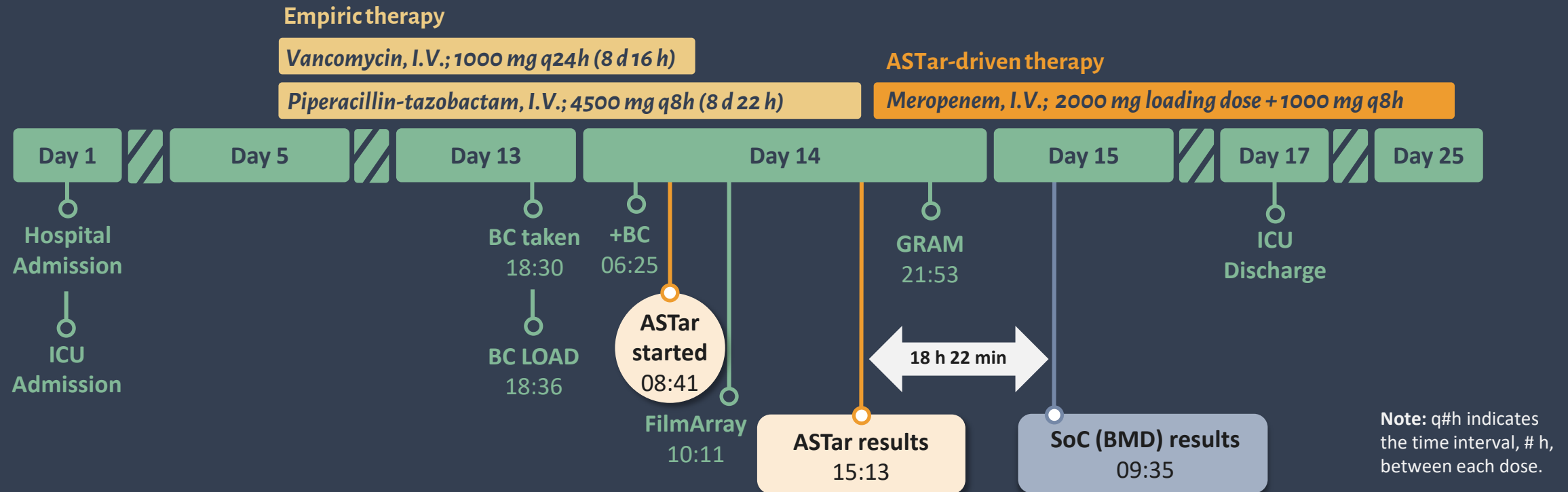
- SOFA score of 9
- SAPS score of 43
- Charleston comorbidity index 3

Clinical timeline

Pathogen ID: *Klebsiella aerogenes*

Phenotype results:
AmpC ind

Genotypic resistance:
None detected



ASTar drove treatment adjustment **>18 hours earlier** than standard of care

AST reports

Pathogen ID:

Klebsiella aerogenes

Phenotype results:

AmpC inducible

Genotypic resistance:

None detected

Empiric therapy:

Piperacillin-tazobactam

Vancomycin

ASTar-driven therapy:

Meropenem

Patient outcomes

- No side effects from antibiotics
- Discharged from the ICU 3 days after starting ASTar-driven therapy
 - » Clinical cure of the infection
 - » Transfer to internal medicine ward
 - » Planned transfer to rehabilitation center

Antibiotic	Breakpoints		ASTar		SoC (BMD)	
	S ≤	R >	MIC	SIR	MIC	SIR
Amikacin	8	8	4	S	≤4	S
Amoxicillin-clavulanic acid	8	8	>32	R	-	-
Aztreonam	1	4	>64	R	-	-
Cefazolin	0.001	4	>16	R	-	-
Cefepime	1	4	2	I	-	-
Cefotaxime	1	1	>128	R	>2	R
Ceftazidime	1	4	>64	R	>128	R
Ceftazidime-avibactam	8	8	4	S	2	S
Ceftolozane-tazobactam	2	2	>32	R	>8	R
Ceftriaxone	1	2	128	R	-	-
Cefuroxime	0.001	8	>64	R	-	-
Ciprofloxacin	0.25	0.5	0.125	S	≤0.25	S
Ertapenem	0.5	0.5	1	R	-	-
Gentamicin	2	2	2	S	-	-
Levofloxacin	0.5	1	0.25	S	≤0.5	S
Meropenem	2	8	0.5	S	≤0.125	S
Meropenem-vaborbactam	8	8	≤0.25	S	-	-
Piperacillin-tazobactam	8	8	>256	R	>64	R
Tobramycin	2	2	2	S	-	-
Trimethoprim-sulfamethoxazole	2	4	0.125	S	≤1	S
Cefoxitin (screen)	8	8	>64	POS		

Patient case conclusions

Conclusions

- ASTar-driven antibiotic treatment was appropriate (correct microbial coverage)
 - » Escalation to Meropenem as advised by an infectious disease specialist
- No antibiotic change driven by conventional BMD, as part of SoC
- ASTar guided treatment adjustment 18.4 hours earlier than SoC. Results in agreement with BMD